

Hyeong-Ohk Fest on Nonlinear PDEs, Collective Dynamics and Finance

Date: 6th-8th June 2024

Location: Seoul National University
(Building 129 Room101)

6th (Thursday)

7th (Friday)

	Chair: Namkwon Kim	Chair: Jaeyoung Yoon
10:00-10:30	Hi Jun Choe (Yonsei University) TBA	Jongmin Han (Kyung Hee University) Turing instability and attractor bifurcation for the 1D Brusselator model
10:30-11:00	Yonghoon Lee (Pusan University) TBA	Jihoon Lee (Chung-Ang University) Existence and Stability of the solutions to the Maxwell-Navier-Stokes equations
11:00-11:30	Tongkeun Chang (Yonsei University) The properties of solution for incompressible Stokes equations and Navier-Stokes equations near boundary	Moonjin Kang (KAIST) From Navier-Stokes to BV solutions of the barotropic Euler equations
11:30-12:00	Dongnam Ko (The Catholic University of Korea) On the interactions of flocking particles with the Stokes flow in an infinite channel	Joerg Wolf (Chung-Ang University) A new condition on the vorticity for partial regularity of a local suitable weak solution to the Navier-Stokes equations
12:00-14:00	Lunch	Lunch
	Chair: Dongnam Ko	Chair: Jane Yoo
14:00-14:30	Minkyu Kwak (Chungnam National University) A new approach to the theory of inertial manifolds	Doheon Kim (Hanyang University) A Lyapunov approach to score-based generative models with multiplicative noise conditioning
14:30-15:00	Youngsam Kwon (Dong-A University) Conditional regularity for the compressible MHD system with inhomogeneous boundary data	Jeongho Kim (Kyung Hee University) Long-time behavior towards viscous-dispersive shock for Navier-Stokes equations of Korteweg type
15:00-15:30	Byungjoon Lee (The Catholic University of Korea) Oscillatory Pattern Capturing Network for Highly Oscillatory Data	Gyoocheol Shim (Ajou University) Optimal Consumption and Investment with Costly Reversible Job-Switching Option
15:30-16:00	Coffee Break	Coffee Break
	Chair: Youngsam Kwon	Chair: Seung Yeal Ha
16:00-16:30	Young-Pil Choi (Yonsei University) Global Cauchy problem for the Euler-Poisson-Navier-Stokes system	Seung-Yeon Cho (Gyeongsang National University) A conservative semi-Lagrangian scheme for the ES-BGK model of the Boltzmann equation
16:30-17:00	Sang-Hyeok Lee (Korea Maritime Institute) Maritime Economics & Deep Neural Networks	Myeong-Su Lee (KAIST) Solving Inverse Problems in Partial Differential Equations using the Unsupervised Legendre-Galerkin Neural Network
17:00-17:30	Hantaek Bae (UNIST) On Decay Rates of Solutions to the Incompressible Navier-Stokes Equations	Hyeong-Ohk Bae (Ajou University) TBA
18:00-20:00		Banquet (Hoam Guest House)